

AI Hackathon:

AI-Driven Contract Lifecycle Management (CLM)

1. Project Vision & Goal

To develop a prototype of an AI-powered platform that automates and enhances the management of legal contracts. The primary goal is to reduce manual effort, minimize human error, accelerate the review process, and provide actionable, legally sound insights from contract data. The system should support both English and Arabic with highly professional drafting capabilities.

2. The Problem

Manual contract management is slow, expensive, and risky. Legal and procurement teams spend countless hours manually reviewing documents, extracting key information, and tracking important dates and obligations. This process is prone to error, leading to missed deadlines, non-compliance with terms, and significant financial or legal exposure. Furthermore, valuable data remains locked within unstructured documents, making strategic analysis of the entire contract portfolio nearly impossible.

3. The AI Solution: High-Level Requirements

Build a web-based application that allows users to upload contracts and uses AI to automatically analyze them. The system should present extracted information, risks, and summaries in a clear, user-friendly dashboard. The solution will be dynamic, allowing for auto-refresh updates based on the industry, subject, or sector selected.

4. Key Features & Functional Requirements (MVP)

A successful hackathon prototype should aim to implement the following core features:

- **Feature 1: Secure Contract Upload & Processing**
 - **Requirement:** Users must be able to upload contract documents (e.g., in .pdf or .docx format) through a web interface.
 - **User Story:** As a legal officer, I want to upload a new vendor agreement so the system can analyze it for me.
- **Feature 2: AI-Powered Data Extraction**
 - **Requirement:** The AI core must automatically identify and extract key pieces of information from the contract text.
 - **Data to Extract:**

- Contracting Parties: Names of all organizations/individuals involved.
- Key Dates: Effective Date, Expiration Date, Renewal Date.
- Governing Law & Jurisdiction: The legal framework governing the contract.
- Key Obligations & Deliverables: Critical duties for each party.
- Financial Terms: Payment amounts, schedules, and penalties.
- **User Story:** As a procurement manager, I need the system to automatically pull all payment deadlines from a contract so I can add them to my calendar.
- **Feature 3: Risk & Compliance Analysis**
 - **Requirement:** The AI should analyze the contract for non-standard clauses, potential risks, or missing information. The system should also provide legal advice, opinions, and comparisons with applicable laws and regulations.
 - **Analysis Points:**
 - Identify unusual or high-risk clauses (e.g., unlimited liability, unusual termination clauses).
 - Flag missing standard clauses (e.g., confidentiality, force majeure).
 - Check for consistency in defined terms.
 - **User Story:** As a compliance officer, I want the AI to flag any clauses that don't match our standard company templates.
- **Feature 4: Intelligent Dashboard & Visualization**
 - **Requirement:** A central dashboard that provides a high-level overview of the contract.
 - **Dashboard Components:**
 - Display a summary of the extracted key data.
 - Show a list of identified risks and alerts.
 - Provide a concise, AI-generated summary of the contract's purpose and scope.
 - Present AI-generated advice, opinions, and legal comparisons.
 - **User Story:** As a manager, I want to see a one-page summary of any contract so I can quickly understand its main points without reading the whole document.
- **Feature 5: Smart Clause Automation (Stretch Goal)**
 - **Requirement:** The solution should be dynamic and update automatically based on the selected industry, subject, or service. Each selection will have a set of default clauses. When a user selects a specific governing law (e.g., Qatar or UK) or jurisdiction, the corresponding paragraphs and clauses will be automatically updated to ensure legal compatibility.
 - **User Story:** As a legal professional, I want to generate a new contract where the clauses are automatically tailored to my industry and the governing law I select.
- **Feature 6: Natural Language Search (Stretch Goal)**
 - **Requirement:** Allow users to ask questions about the contract in plain English.
 - **Example Queries:** "What is the penalty for late delivery?", "When does this agreement expire?", "Who is responsible for data security?"
 - **User Story:** As a project manager, I want to ask the system specific questions about a contract's terms without having to search through the document manually.

5. Technical & Non-Functional Requirements

- **UI/UX:** The application must be intuitive, clean, and easy to navigate. The focus should be on clearly presenting the AI-generated insights. The system must be professional and support both English and Arabic.
- **Technology Stack (Suggested):**
 - **Backend:** Python (with a web framework like Flask or FastAPI) is recommended due to its strong AI/ML libraries.
 - **Frontend:** A modern JavaScript framework (like React, Vue) or even simple HTML/CSS with JavaScript for a rapid prototype.
 - **AI/ML:** Utilize Large Language Models (LLMs) via APIs (e.g., Gemini) for summarization and Q&A. For data extraction, consider fine-tuning smaller models or using advanced prompting techniques with LLMs.
- **Performance:** The analysis of a standard contract (10-20 pages) should be completed in a reasonable time frame (e.g., under 60 seconds).
- **Data Security:** While this is a prototype, teams should mention how they would handle sensitive contract data in a real-world application (e.g., encryption, access controls).

6. Success & Judging Criteria

Teams will be evaluated based on:

- **Core Functionality:** How well was the AI-powered data extraction and risk analysis implemented?
- **Innovation & "Wow" Factor:** Does the solution offer a creative approach or a particularly insightful feature (e.g., the natural language search or the smart clause automation)?
- **User Experience (UX):** How effective and easy-to-use is the final prototype? Is the dashboard clear and actionable?
- **Business Impact:** How well does the prototype solve the stated problem and demonstrate value for an organization like QDB?